SUPPORTING ADULT GUIDANCE

The angle bisector

Constructions, Loci and Bearings · Lesson 4. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This lesson introduces the angle bisector as both a **construction** and a **locus** — the set of all points equidistant from the two arms of an angle. It deliberately parallels Lesson 3 (the perpendicular bisector, equidistant from two points), reinforcing the powerful locus idea. Learners discover the line by collecting equal-distance points, construct it exactly with equal arcs, meet the fold symmetry that explains it, and apply it to fit a circle inside the angle.

Driving Question: *How can I split any angle into two exactly equal halves?*

The mathematics, in plain terms

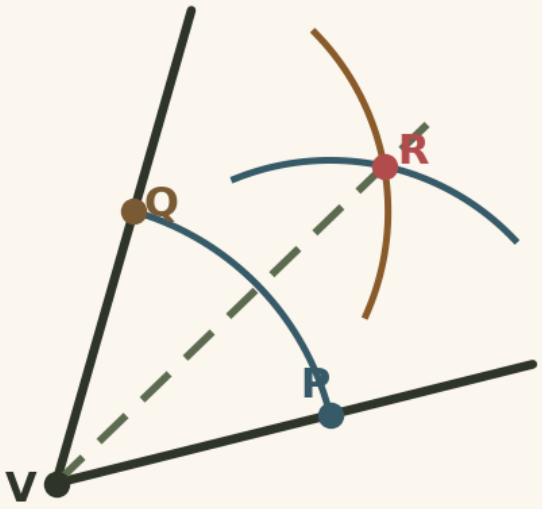
The distance from a point to a line is measured along the perpendicular. A point equidistant from both arms of an angle lies on the angle bisector, the line from the vertex that halves the angle. The construction works by symmetry: an arc from the vertex marks two points P and Q equally far along the arms, and equal arcs from P and Q meet on the line of symmetry of the figure — the bisector. A useful consequence: the centre of any circle that fits snugly against both arms lies on the bisector, because it must be the same distance (the radius) from each arm.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Construct the bisector of an angle

Construct the bisector of the angle at V.

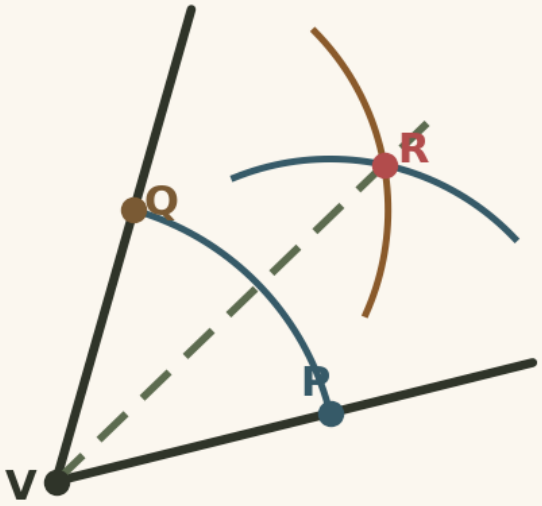


1. From V, swing an arc cutting both arms at P and Q.	arc from V
2. Swing an equal arc from P.	arc from P
3. Keep the opening; swing an equal arc from Q, crossing at R.	arc from Q
4. Draw the line from V through R.	join V–R

V–R is the angle bisector: it splits the angle into two equal halves, and every point on it is the same distance from both arms.

Example 2 — Halving an angle

The bisector splits the angle into two equal halves. If the whole angle is 80°, how big is each half?



1. Construct the bisector V–R (as above).	line V–R
2. It divides the 80° angle into two equal parts.	80 ÷ 2
3. Each half is 40°.	= 40°

Each half is 40° — the bisector halves the angle exactly, with no protractor.

Locus first, then the exact construction

As in Lesson 3, let learners *discover* the line before constructing it: find several points equidistant from both arms and watch them line up. Meeting the bisector as 'all the points that…' keeps the locus idea alive. The fold interactive then gives the reason it halves the angle — the picture is symmetrical about the bisector. Invite the learner to describe the symmetry before naming the line.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Find a path equally far from two roads meeting at a junction, and notice such points line up.	~9 min
Movement	Drag a point to collect points equidistant from both arms; the bisector emerges as their locus.	~13 min
Reflection	Construct the bisector with equal arcs, and fold the angle to see the symmetry that halves it.	~10 min

Creativity	Slide a centre along the bisector and watch its circle touch both arms.	~9 min
Rest	A short pause to notice how this lesson rhymes with the last.	~2 min
Nutrition	Name what the learning feeds: patterns across ideas, and symmetry as a reason for exactness.	~2 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

“The bisector is any line through the vertex that looks central.”

It must be equidistant from both arms. Ask the learner to test points along their line — are they really the same perpendicular distance from each arm?

“The two arcs from P and Q need different radii.”

They must be equal, so the crossing is the same distance from P and Q. Ask: 'If the crossing must sit on the line of symmetry, what must the two arcs be?'

“You measure the distance to an arm along any direction.”

Distance to a line means the perpendicular distance. Ask the learner to drop a right-angle from the point to each arm.

Confusing the angle bisector with the perpendicular bisector.

Both are loci of equidistant points, but from different things: perpendicular bisector from two points, angle bisector from two lines. Ask which two things the points are equidistant from.

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: equal-distance compass arcs (Lesson 1) and the perpendicular bisector (Lesson 3). A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner draws a line that halves the angle by eye, or is unsure why equal arcs matter. The aim is a question that hands the reasoning back.

- “You found one point equally far from both arms. Can you find another, further from the vertex?”
- “What do all your equal-distance points have in common?”
- “Both arcs from P and Q were the same size — so where must their crossing sit relative to the two arms?”
- “If you folded the angle along your line, would one arm land exactly on the other? How do you know?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Do we need a protractor?

No. The equal-arc construction halves the angle exactly; the equal halves are a consequence of the symmetry, not something you measure.

Why does the arc from the vertex have to cut both arms?

It marks two points, P and Q, that are equally far along the arms. Those equal distances are what make the rest of the construction symmetrical.

How is this different from last lesson's perpendicular bisector?

Both are loci of equidistant points. The perpendicular bisector is equidistant from two points; the angle bisector is equidistant from two lines (the arms).

What makes good evidence here?

A clear construction with arcs showing, the vertex joined to the crossing, and a sentence explaining that every point on the line is the same distance from both arms.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like

Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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